Computer Assignment 1
Univariate volatility models

Introduction

Here you will implement the various univariate volatility models we have discussed and also use (filtered) historical simulation for estimating VaR. These methods are used for measuring market risk, estimate parameters for portfolio selection and more.

Important: You are not allowed to use existing toolboxes or packages for estimating GARCH-parameters, you need to write this code yourselves. AI-generated code is <u>not</u> allowed.

1. Daily volatility forecast using MA and EWMA

Perform a daily volatility forecast using MA and EWMA with a 60 day estimation window, over the time period 2022-01-01 -> 2025-12-31 for the S&P 500 index (^SPX). Plot the daily forecasts in the same plot. Plot also the log-returns in a separate plot below.

Discussion: What is the main reason for the differences in these two forecasts.

2. Daily volatility forecast using GARCH

Fit a GARCH(1,1)-model using Nasdaq return data for the time period 2012-01-01 -> 2021-12-31 and then perform a daily volatility forecast as in 1, with forecasting period: 2022-01-01 -> 2025-12-31. Plot your daily forecast in the same plot as in 1.

Discussion: What is the main reason for the difference between the GARCH and the EWMA forecast.

3. GARCH model diagnostics

Analyze the GARCH-model and perform model diagnostics:

a) What are the GARCH-parameters and what is the unconditional <u>volatility</u>?

b) Plot the ACF (for lags 1 to 200) for the returns and the squared returns (use a stem-plot).
c) Plot the ACF (for lags 1 to 200) for the residuals and the squared residuals (use a stem-plot).
d) Plot a histogram (using 100 bins) of the residuals together with a normal pdf (same mean and std).
e) QQ-plot the residuals against normal- and student t quantiles. Approximately which degrees of freedom provides the best fit to the left tail?

Discussion: What does b) and c) tell you? What does d), e) and f) tell you?

4. VaR forecasts using historical simulation

Use backtesting to compare daily $VaR_{98\%}$-forecasts for the period 2022-01-01 -> 2025-12-31 using two versions of basic historical simulation with rolling estimation windows of length 500 and 1000 days respectively.

a) Plot the VaR-forecasts in the same plot as the (negative) returns
b) Calculate the number of violations and the violation ratios

Discussion: What are the pros and cons of using these methods?

5. VaR forecasts using GARCH and FHS

Repeat the backtest procedure in 4 but now based on the GARCH-model from 2. The two models to backtest are:

- GARCH with the assumption of normally distributed conditional returns
- One-period filtered historical simulation (FHS)

You do not need to recalibrate the GARCH-model during the backtest period.

a) Plot the VaR-forecasts in the same plot as in 4
b) Calculate the number of violations and the violation ratios
c) Determine the 95% acceptance interval for the violation ratio under the assumption that the VaR forecasts are correct. Is any model rejected (include also the HS models)?

Discussion: What are the pros and cons of using these methods?

Grading

These are the requirements for the grades 3, 4 and 5:

Problem	3	4	5
1	All	All	All
2	All	All	All
3	All	All	All
4	-	All	All
5	-	-	All